

The exact calculations behind Damodaran's data.

Every formula the datasets use — quoted from Damodaran's own "Variables & FAQ" sheets and variable-definitions page, verified against the numbers we ingested. Covers the value-weighting rule, the R&D and lease adjustments, margins, financing flows, multiples, FCFE — and **where the EPS calculation actually lives**.

1 The aggregation rule & building blocks

2 Capitalizing R&D — the centerpiece

3 Capitalizing leases & the margins

4 The R&D dataset

5 Financing flows (+ a real contradiction)

6 Valuation multiples & where EPS lives

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8 Sources & caveats

Primary source: the "Variables & FAQ" sheet inside each Damodaran .xls (Jan-2026), plus his variable-definitions page. Formulas in quotes are his exact wording.

1 The one rule, and the building blocks

Read this first — it governs every single number in every file.

■ The aggregation rule: value-weighted, never an average of ratios

For any industry ratio, Damodaran **sums the numerator across all firms, sums the denominator, then divides**. His FAQ repeats the phrase "aggregated ... as a percent of aggregated ..." for every metric.

$$\text{Industry ratio} = \Sigma(\text{numerator, all firms}) / \Sigma(\text{denominator, all firms}) \text{ e.g. Gross Margin} = \Sigma(\text{gross profit}) / \Sigma(\text{revenue})$$

Consequence: each figure is a **value-weighted composite dominated by the largest firms** — the "industry as one merged company," not the typical firm. (He even labels the count column's rationale "*Law of large numbers?*".)

■ The building blocks (defined once, reused everywhere)

TERM	DAMODARAN'S DEFINITION
Effective tax rate	taxes paid / taxable income (as reported to stockholders)
EBIT	operating income (revenue – COGS – SG&A – R&D – D&A)
EBITDA	EBIT + depreciation & amortization
Invested Capital	BV of equity + BV of debt – cash
Enterprise Value (EV)	market value of equity + debt – cash
Net Debt	total debt (incl. leases) – cash & marketable securities

WHY TWO OF THESE GET "ADJUSTED"

Damodaran argues two things are **mis-classified by accounting**: **R&D** (really an investment, but expensed) and **operating leases** (really debt, but hidden). He re-computes operating income and invested capital to fix both — the next two sections. Every "adjusted" column in the margin file flows from these two corrections.

2 Capitalizing R&D — the centerpiece

This is the whole basis of the margin-distortion thesis. Here is exactly how he does it.

GAAP expenses R&D immediately. Damodaran instead treats it as a capital asset amortized over an **industry-specific life** (longer for pharma, shorter for software — matching how long the research stays commercially valuable). His FAQ states a **5-year straight-line** default.

■ Step 1 — build the "Research Asset" (the unamortized R&D)

With a 5-year life, this year's R&D counts fully; last year's is 4/5 remaining; and so on:

$$\text{Research Asset} = \text{R\&D}_0 + \frac{4}{5} \cdot \text{R\&D}_{-1} + \frac{3}{5} \cdot \text{R\&D}_{-2} + \frac{2}{5} \cdot \text{R\&D}_{-3} + \frac{1}{5} \cdot \text{R\&D}_{-4}$$
 (this is the "R&D Capitalized" column; "R&D 1–5 years ago" are the inputs)

■ Worked example (illustrative)

YEAR	R&D SPENT	FRACTION UNAMORTIZED (5-YR)	IN THE ASSET
This year	100	5/5	100
1 year ago	90	4/5	72
2 years ago	80	3/5	48
3 years ago	70	2/5	28
4 years ago	60	1/5	12
Research Asset			= 260

■ Step 2 — re-state operating income (his exact FAQ formula)

R&D FAQ: "R&D adjustment to operating income = Operating Income + Current year's R&D exp – Amortization of R&D asset."

$$\text{Adjusted EBIT} = \text{EBIT} + \text{R\&D}(\text{current}) - \text{R\&D amortization}$$
 amortization $\approx \frac{1}{5}$ of each of the last five years' R&D ($\approx$ the trailing average)

THE MECHANISM THAT EXPLAINS OUR RESULT

The reported $\rightarrow$ adjusted **gap** = **(current R&D – amortization) / revenue**. If R&D is **growing**, current > amortization $\rightarrow$ the adjustment **raises** the margin (in our data, Software 0.330 $\rightarrow$ 0.344). If R&D is large but **flat/mature**, amortization of the big accumulated asset $\approx$ current $\rightarrow$ little or negative change (Pharma 0.295 $\rightarrow$ 0.272). That's precisely why the distortion concentrates in the *high-R&D-growth* tail.

Step 3 (used elsewhere): the Research Asset is also **added to invested capital**, which lowers measured Return on Capital — the Capitalized R&D as % of Invested Capital column.

3 Capitalizing leases, and every margin column

The second adjustment, then the full margin.xls dictionary with exact formulas.

■ The lease adjustment (his exact FAQ formula)

Margin FAQ: "Lease adjustment to operating income = Operating Income + Current year's lease exp – Lease Asset Depreciation. Lease expenses are financing, not operating expenses. Hence we add them back."

Operating leases are treated as debt (a lease liability = present value of future commitments), which creates a leased "asset" that is depreciated. Note: after ASC 842 / IFRS 16 (~2019) leases already sit on the balance sheet, shrinking this adjustment — which is why cross-year comparisons need care.

■ margin.xls — every column, exactly

COLUMN	FORMULA (AGGREGATED)
Gross Margin	$\Sigma \text{ gross profit} / \Sigma \text{ revenue}$
Net Margin	$\Sigma \text{ net income} / \Sigma \text{ revenue}$
Pre-tax Unadjusted Operating Margin	$\Sigma \text{ EBIT} / \Sigma \text{ revenue}$ ← the "reported" margin
After-tax Unadjusted Op. Margin	$\Sigma \text{ EBIT}(1-t) / \Sigma \text{ revenue}$
Pre-tax Lease-adjusted Margin	$\Sigma (\text{EBIT} + \text{lease add-back}) / \Sigma \text{ revenue}$
Pre-tax Lease & R&D-adj Margin	$\Sigma (\text{EBIT} + \text{lease} + \text{R\&D} - \text{R\&D amort.}) / \Sigma \text{ revenue}$ ← "adjusted"
EBITDA/Sales	$\Sigma \text{ EBITDA} / \Sigma \text{ revenue}$
EBITDASG&A/Sales	$\Sigma (\text{EBITDA} + \text{SG\&A}) / \Sigma \text{ revenue}$
COGS/Sales · R&D/Sales · SG&A/Sales	$\Sigma (\text{that expense}) / \Sigma \text{ revenue}$
Stock-Based Comp/Sales · Lease Exp/Sales	$\Sigma (\text{that expense}) / \Sigma \text{ revenue}$

THE THESIS IN ONE LINE OF ALGEBRA

"Adjusted" – "reported" operating margin = $(\text{current R\&D} - \text{R\&D amortization} + \text{lease add-back}) / \text{revenue}$.
 Ranking industries by the *reported* line therefore understates exactly the R&D- and lease-heavy ones — the gap our project measures.

4 The R&D and financing-flow files

R&D.xls and finflows.xls — including a definition that contradicts its own data.

■ R&D.xls

COLUMN	FORMULA / MEANING
R&D Capitalized	the Research Asset (§2), 5-yr straight-line, aggregated
Capitalized R&D as % of Invested Capital	Research Asset / (Invested Capital incl. that asset)
R&D – LTM	Σ trailing-12-month R&D
Current R&D as % of Revenue	Σ R&D / Σ revenue
R&D, 1–5 years ago	Σ R&D in each prior year (inputs to the asset)
CAGR in R&D, 5y	$(R\&D_0 / R\&D_{-5})^{(1/5)} - 1$

■ finflows.xls — aggregate cash-flow-statement dollars (trailing 12 months)

COLUMN	FORMULA / MEANING
Dividends · Buybacks · Equity Issuance	Σ of each, across firms
Net Equity Change	Equity Issuance – Buybacks (see box)
Net Equity Change % of Book Equity	Net Equity Change / Σ book value of equity
Debt Raised · Debt Repaid	Σ of each
Net Change in Debt	Debt Raised – Debt Repaid
...% of Total Debt	Net debt change / Σ total debt
Change in Lease Debt	Δ in accountant-computed lease obligations

A VERIFIED CONTRADICTION — TRUST THE DATA, NOT THE NOTE

The finflows FAQ says "Net Equity Change = Stock Issuances – Buybacks – **Dividends**." But the published numbers do **not** subtract dividends. We checked row-by-row: Advertising = $1996.0 - 2180.4 = -184.3$ (the value in the file); subtracting dividends would give -755.4 . So the real formula is **Issuance – Buybacks**. We document the true one and compute "net payout including dividends" separately when needed.

5 Valuation multiples — and the EPS answer

psdata.xls & pedata.xls, plus a direct answer to "where is the EPS calculation?"

COLUMN	FORMULA (DAMODARAN'S WORDING)
PS (Price/Sales)	$\Sigma \text{ market cap} / \Sigma \text{ revenue}$
EV/Sales	$\Sigma \text{ EV} / \Sigma \text{ revenue}$ (EV = Mkt Cap + Debt – Cash)
Current PE	avg of (price per share / EPS, latest year), money-makers only
Trailing PE	avg of (price / trailing-12m EPS), money-makers only
Forward PE	avg of (price / expected next-4-quarter EPS), money-makers
Aggregate Mkt Cap / Net Income	$\Sigma \text{ market cap} / \Sigma \text{ net income}$ (value-weighted PE)
Expected growth, next 5y	avg analyst-estimated EPS growth
PEG Ratio	Current PE / expected 5-yr EPS growth

"WHERE IS THE EPS CALCULATION?" — THREE PLACES, NONE OF THEM A STANDALONE COLUMN

1 · Inside PE. Damodaran doesn't publish an "industry EPS." He computes $\text{PE} = \text{price per share} \div \text{EPS}$ for each firm, then averages across money-making firms — so EPS is the denominator of every PE column ($\text{EPS} = \text{net income} \div \text{shares}$).

2 · Money-losers are dropped. His FAQ warns simple-average PEs exclude negative-EPS firms ("that will create a bias"); the `Aggregate Mkt Cap / Net Income` columns are the value-weighted alternative that keeps them.

3 · EPS growth is a separate file. Expected EPS growth in pedata is analyst-based; the *fundamental* EPS growth formula lives in `fundgr.xls` — see §6. (We loaded its operating-income sibling, `fundgrEB.xls`.)

6 Payout, FCFE & the two growth formulas

divfcfe.xls and the fundamental-growth files — the second answers the EPS-growth question.

■ divfcfe.xls — cash returned vs. cash that could be returned

COLUMN	FORMULA (DAMODARAN'S WORDING)
Payout	$\Sigma \text{ dividends} / \Sigma \text{ net income}$
Cash Return	$\Sigma (\text{dividends} + \text{buybacks}) / \Sigma \text{ net income}$
FCFE, before debt cash flows	$\Sigma (\text{net income} + \text{depreciation} - \text{capex incl. R\&D \& acquisitions})$
FCFE, after debt cash flows	$\dots + \Sigma (\text{new debt raised} - \text{debt repaid})$
Net Cash Returns as % of FCFE	$(\text{div} + \text{buybacks} - \text{net issuance}) / \text{FCFE}$

FCFE is "potential dividends" — cash left after taxes, reinvestment and debt flows (Damodaran calls the before-debt version "*Owner's earnings*"). Net-cash-returned > **100%** of FCFE flags an industry paying out more than it generates.

■ The two fundamental-growth formulas

Growth = how much you reinvest × how well it earns. There's an **equity** version (per-share earnings) and an **operating** version (EBIT):

$\text{EPS growth (fundgr.xls)} = \text{Retention Ratio} \times \text{ROE}$ $\text{Retention} = 1 - \text{payout}$; $\text{ROE} = \text{net income} / \text{BV equity}$
 $\text{EBIT growth (fundgrEB.xls)} = \text{Reinvestment Rate} \times \text{ROC}$ ← the file we loaded
 $\text{Reinvestment} = (\text{net capex incl. R\&D \& acq.} + \Delta \text{ non-cash WC}) / \text{EBIT}(1-t)$ $\text{ROC} = \text{EBIT}(1-t) / \text{Invested Capital}$

DIRECT ANSWER TO THE EPS-GROWTH QUESTION

The **EPS growth calculation is Retention Ratio × ROE**, published in fundgr.xls. Our project loaded the operating-income analogue, fundgrEB.xls (Reinvestment × ROC), because the thesis is about operating margins, not per-share earnings. Both are the same idea — *reinvestment × return* — one for equity, one for the whole firm.

7 Caveats baked into the calculations

What to keep in mind when you use any of the formulas above.

- **They're estimates where it counts.** The R&D amortizable life, the lease discount rate, and the effective-vs-marginal tax choice are Damodaran's assumptions — the capitalized-R&D and adjusted margins are *model outputs*, not GAAP figures.
- **Value-weighted ≠ typical firm.** Every ratio is dominated by the industry's mega-caps. Nothing here supports a claim about an individual company.
- **Negative / near-zero denominators distort ratios.** Book equity can go negative after years of buybacks; net income can be negative — so %-of-book and PE columns blow up. He publishes money-makers-only PE and "without financials" totals for this reason; we winsorise downstream.
- **Financials are special.** Banks/insurers have no comparable "revenue" or "operating margin," so they're excluded from margin/ROC computations.
- **One snapshot.** All trailing-12-months through ~Q3 of the prior year — a single point in the cycle, not a trend. The "N years ago" columns are computed on *today's* survivors (survivorship bias).
- **Definitions can lag the data** — the Net Equity Change contradiction (§4) is the proof; always sanity-check a formula against a couple of rows.

HOW THIS FEEDS THE PROJECT

Our pipeline reads these columns as-is into a database and computes two things on top: the **reported-vs-adjusted margin gap** (from the two adjustments in §2–3) and whether the market's **EV/Sales** (§5) tracks the reported or adjusted margin. Everything upstream is Damodaran's arithmetic; we don't re-derive it — we *compare* its outputs.

8 Sources (all primary, all free)

Every formula above is quoted or derived from these Damodaran sources.

■ Primary — the exact column definitions

- The "**Variables & FAQ**" sheet inside each `.xls` — the authoritative, per-file definitions (margin, R&D, finflows, psdata, pedata, divfcfe, fundgrEB). Quoted verbatim throughout.
- [Master variable definitions](#) — `datafile/variable.htm`

■ R&D & lease capitalization

- [Research & Development Expenses \(Accounting Primer\)](#)
- [Measuring Earnings \(Ch. 9\)](#) · [Valuing Companies with Intangible Assets](#)

■ Growth, PE & PEG

- [The Fundamental Determinants of Growth](#) (Retention × ROE)
- [Price-Earnings Ratio](#) · [PEG Ratios](#)

VERIFIED, NOT JUST CITED

Two formulas in this document were checked against the raw numbers we ingested: the **Net Equity Change** contradiction (§4, confirmed = Issuance – Buybacks) and the **R&D/Sales** vs. EBITDAR&D/Sales column mapping. The rest are quoted from Damodaran's own FAQ sheets.